HW #9.

Prob. 1. Solve

(1) $\frac{dy}{dx} = y - x.$

(2) $y(0) = 1.$ using the method:

(3) $y_{m+1} + \frac{3}{2} y_m - 3y_{m-1} + \frac{1}{2} y_{m-2} = 3h\, f_m.$

This is the exact problem with previous HW #8 Problem 2 I think. So, I'll be uploading the solution from it on the next starting page:

Prob. 2. Solve

(1) $\frac{dy}{dx} = y - x.$

(2) $y(0) = 1.$ using the ↓ method

(3) $y_{i+1} + \frac{3}{2} y_i - 3 y_{i-1} + \frac{1}{2} y_{i-2} = 3h f_i$

Sol-n.

(1) $\frac{dy}{dx} = y - x$

$y' - y = -x. \quad \Big| \cdot e^{-x}$

$y' \cdot e^{-x} - y \cdot e^{-x} = -x e^{-x}$

$\frac{d}{dx}\left(y \cdot e^{-x}\right) = -x \cdot e^{-x}$

$\int d\left(y \cdot e^{-x}\right) = \int -x e^{-x} dx + C.$

$y e^{-x} = \int \underbrace{x}_{u} d\underbrace{(e^{-x})}_{v} + C \Rightarrow$

$y \cdot e^{-x} = x e^{-x} - \int e^{-x} dx + C.$

$y \cdot e^{-x} = x e^{-x} + e^{-x} + C. \quad \Big| \cdot e^{x}$

(4) $\boxed{y = x + 1 + C \cdot e^{x}}$ using the IV in (2):

$y(0) = 1 = 0 + 1 + c \cdot e$

$ce^0 = 0 \Rightarrow \boxed{c = 0}$

(6) $\boxed{y(x) = x + 1}$

Note the solution exists because

(7) $f(x,y) = y - x \in C(\mathbb{R}^2) =$ cont-us on $\mathbb{R}^2$.

and unique because

(8) $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(y - x) = 1 \in C(\mathbb{R}^2) =$ con-t on $\mathbb{R}^2$

Using the L-M method in (3):

(9) $y_{i+1} = \frac{-3y_i}{2} + 3y_{i-1} - \frac{y_{i-2}}{2} + 3h \cdot f(x_i, y_i)$

$\Rightarrow$

(10) $y_{i+1} = \frac{-3y_i}{2} + 3h(y_i - x_i) + 3y_{i-1} - \frac{y_{i-2}}{2}$

say $\boxed{i = 2}$

(11) $y_3 = \frac{-3y_2}{2} + 3h(\overset{?}{y_2 - x_2}) + 3\overset{?}{y_1} - \frac{y_0}{2}$. ✓

x_0 y_0 x_1 y_1 x_2 y_2 Exact y_2

To find y_3 we need extra y_1, x_2, y_2 values which we dont know. For the we are allowed to use the exact sol-n (6).

(6) $\boxed{y(x) = x+1}$ $x_0 = 0$,

(12) $\begin{cases} x_1 = x_0 + h. \\ y_1 = y(x_1) = x_1 + 1 \end{cases}$

But to work on (12) we need h!

Since (3) is explicit L.M. method, it does have cond-n on h, which we provided in Pr.1.

(13) $h < \frac{2}{\lambda}$, $\lambda = \max\left\{ \left|\frac{\partial f}{\partial y}\right| ; (x,y) \in D(f) \right\}$

$y(7)$

$\lambda = \max\left\{ \left|\frac{\partial}{\partial y}(y-x)\right| ; (x,y) \in \mathbb{R}^2 \right\} = 1.$

So $h < \frac{2}{\lambda} = \frac{2}{1} = 2 \Rightarrow$

(14) $\boxed{h < 2}$.

To check how choice of h influences the

numerical sol-s, I'm going to write a code for

$$h = \frac{1}{2^3}, \frac{1}{2^2}, \frac{1}{2} < 2.$$

...

Problem 2. Use Picard's iteration to solve:

(i) $\frac{dy}{dx} = xy$ (1)

(2) $y(0) = 1 \qquad x \in [0,1]$

Solution:

$$dy = xy\,dx \quad \Big\{ \text{BS} \int_{x_0}^{x} \ldots ds \quad \Rightarrow$$

$$\int_{x_0}^{x} dy = \int_{x_0}^{x} s y\, ds$$

(3) $$\boxed{y = y_0 + \int_{x_0}^{x} s y\, ds}$$ Picard Iteration formula:

$$y_1 = y_0 + \int_{x_0}^{x} s y_0\, ds$$

$$y_2 = y_0 + \int_{x_0}^{x} s y_1\, ds$$

$$y_3 = y_0 + \int_{x_0}^{x} s y_2\, ds$$

$$\ldots$$

So the arithmetics are

$$y_1 = y_0 + \int_{x_0}^{x} s y_0 \, ds = 1 + \int_{x_0=0}^{x} s \, ds = 1 + \frac{s^2}{2}\Big|_{x_0=0}^{x} = 1 + \frac{x^2}{2}$$

$$y_2 = 1 + \int_0^x s \cdot \left(1 + \frac{s^2}{2}\right) ds = 1 + \int_0^x s + \frac{s^3}{2} \, ds = 1 + \frac{s^2}{2} + \frac{s^{3+1}}{2\cdot(3+1)}\Big|_0^x$$

$$= 1 + \frac{x^2}{2} + \frac{x^4}{2\cdot 4}$$

$$y_3 = 1 + \int_0^x s \cdot \left(1 + \frac{s^2}{2} + \frac{s^4}{2\cdot 4}\right) ds = 1 + \frac{s^2}{2} + \frac{s^4}{2\cdot 4} + \frac{s^6}{2\cdot 4\cdot 6}\Big|_0^x =$$

$$= 1 + \frac{x^2}{2} + \frac{x^6}{2\cdot 4\cdot 6} + \frac{x^4}{2\cdot 4},$$ So by pattern

$$y_4 = 1 + \frac{x^2}{2} + \frac{x^4}{2\cdot 4} + \frac{x^6}{2\cdot 4\cdot 6} + \frac{x^8}{2\cdot 4\cdot 6\cdot 8}$$ & so

$$y_n = 1 + \frac{x^2}{2} + \frac{x^4}{2\cdot 4} + \frac{x^6}{2\cdot 4\cdot 6} + \dots + \frac{(x^2)^n}{2^n \cdot n!} = \frac{\left(\frac{x^2}{2}\right)^n}{n!} \approx e^{\frac{x^2}{2}}$$

So $y(x) = e^{\frac{x^2}{2}}$.

(ii)

(1) $\frac{dy}{dx} = 2(y+1)$

(2) $y(0) = 0 \qquad x \in [0,1].$

Sol-n:

$dy = (2(y+1)\,dx \qquad \int_{x_0=0}^{x} \ldots ds$ of BS:

$$\int_0^x dy = \int_0^x 2(y+1)\,ds$$

(3) $\boxed{y(x) = y_0 + \int_0^x 2(y+1)\,ds}$ So

$$y_1 = y_0 + \int_0^x 2(y_0+1)\,ds = 0 + \int_0^x 2\,ds = 2x$$

$$y_2 = y_0 + \int_0^x 2(y_1+1)\,ds = 0 + \int_0^x 2(2s+1)\,ds = 2x^2 + 2x$$

$$y_3 = \int_0^x 2\cdot(2s^2+2s+1)\,ds = 2\cdot 2\cdot \frac{s^3}{3}\Big|_0^x + 2s^2\Big|_0^x + 2s\Big|_0^x$$

$$= 2x^2 + \frac{4}{3}x^3 + 2x$$

$$y_4 = \int_0^x 2\left(2s^2 + \frac{4}{3}s^3 + 1\right)ds = 2\left(s^2 + \frac{2s^3}{3} + \frac{4}{3\cdot 4}s^4 + s\right)\Big|_0^x$$

$$= 2\left(x + x^2 + \frac{2x^3}{3} + \frac{1}{3}x^4\right) \quad \ldots$$

I, to be honest, could not fell the pattern for picking the Taylor series, so I stopped at y_4. But I believe this is the process for the Picard's iteration.

Prob 3. Exact discrete solution of the IVP

(1) $y' = -5y$

(2) $y(0) = 1$

3 points given so the given Runge Kutta method is order 2.

Exact sol-n:

$$\frac{dy}{dx} = -5y$$

$$\int \frac{dy}{y} = \int -5 \, dx + \ln C.$$

$$\ln|y| = \ln e^{-5x} + \ln C \Rightarrow \ln|y| = \ln(e^{-5x} \cdot C)$$

$$y = e^{-5x} \cdot C \overset{IV}{\Rightarrow} y(0) = 1 = e^{-5 \cdot 0} \cdot C \Rightarrow \boxed{C = 1}$$

So,

(3) $\boxed{y(x) = e^{-5x}}$. So like we covered in class the exact solution for the given $\boxed{y' = -\lambda y, \; y(x_0) = y_0}$ is as

(4) $\boxed{y_m(x) = A \cdot e^{-\lambda x},}$ so

$y_m(x) = Ae^{-5x}$ & by $y(0) = 1 \Rightarrow A = 1 \Rightarrow$

(5) $\boxed{y_m(x) = e^{-5x}}$ which matches exact sol-n in (3).